

Restoring the Truncated Band: Geometry-Aware Spectral Operators and a Direct Measurement of Geometric Forgetting

Sehaj R. Singh

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Abstract

Fourier Neural Operators (FNO) learn mappings between function spaces with $O(N \log N)$ cost, but their global mixing channel is *structurally band-limited*: mode truncation discards every Fourier mode above the cut, so sharp boundary content and long-range boundary interactions must either be re-synthesized by pointwise nonlinearities or be lost. We give a precise statement of this truncation band-limitation, and show that a multiplicative, field-valued modulation of the spectral weights—derived from a signed distance function (SDF) of the domain—restores the missing band through *spectral convolution*: the product of a low-band latent with a kinked modulation field regenerates high-frequency content at exactly the wall locations where the truncated operator is wrong. The resulting architecture, AGF-NO, reduces to an exact FNO at initialization (zero-gated), preserving the standard training dynamics until the gates open. We then do what the operator-learning literature has consistently failed to do for “geometric forgetting”: *measure* it. A closed-form linear probe on each block’s latent features shows that decodable boundary information in a depth- L FNO collapses monotonically (R^2 0.8961 $\rightarrow$ 0.4716 over four blocks), while AGF-NO’s re-injection measurably slows the decay (R^2 0.9342 $\rightarrow$ 0.5913). On an obstacle-Darcy benchmark with a genuine multigrid-preconditioned solver as ground truth, the mechanism—not extra parameters, input channels, or loss reweighting—accounts for the gain: a gates-frozen capacity control matches the FNO baseline (0.9714 $\times$ its global error) while the full mechanism beats its own frozen self by 33% in global and 55% in near-boundary error. Multi-seed ($n=3$): 0.0805 ± 0.0019 vs. 0.1232 ± 0.0012 relative L^2 . The effect compounds with depth: at 16 blocks the vanilla FNO collapses outright (relative error 1.0053—worse than predicting the mean, with the probe showing *total* geometric collapse, $R^2 = 0.0003$) while AGF-NO remains depth-stable (0.0841 ± 0.0012 over three seeds, probe plateaued at $R^2 \approx 0.6327$). The collapse replicates: a three-seed re-run of the sweep reproduces it with a 11.9592 $\times$ error gap and zero overlap across seeds. It also lifts: on 3-D obstacle Darcy (spheres and solid tori, multiply connected, float64 ground truth) the same collapse-plateau pair reappears at depth 16 (FNO 1.0082 vs. AGF-NO 0.1127; probe R^2 0.0204 vs. 0.4940), establishing the phenomenon as dimension-independent. We also locate the mechanism’s boundary honestly: on a second PDE family (advection-diffusion past fixed-temperature obstacles) whose difficulty is not wall-anchored, the same architecture with the same control shows only a 2% effect—a near-null result we report and hypothesize about, not hide. Code, checkpoints, per-seed results, and one-click GPU reruns are public.

1 Introduction

Neural operators that learn in the frequency domain—most prominently the Fourier Neural Operator (Li et al., 2021)—achieve speedups of orders of magnitude over classical solvers while generalizing across discretizations (Brandstetter et al., 2022; Takamoto et al., 2022). Their central

computational primitive, the spectral convolution, applies a learnable complex weight to the low-frequency modes of the latent field and discards the rest. The truncation is presented as a benign regularization; the folklore justification is “solutions are smooth.”

For problems whose *difficulty lives at the boundary*, the justification is false. Fluid-thermal boundary layers, wall discontinuities of heterogeneous media, and shock-adjacent structure are precisely the high-frequency, geometry-coupled content of the solution. If the operator’s only global mixing channel is band-limited, that content must survive—if it survives at all—through pointwise nonlinearities acting on low-band inputs, a channel with no access to where the walls are unless the walls are re-supplied at every depth.

This article is written as an analysis-and-progress contribution rather than a benchmark paper: it (i) *diagnoses* a structural failure mode of spectral operators, (ii) supplies a *measurement instrument* (a controlled linear-geometry probe with null and sanity controls) that makes the failure quantifiable block by block, (iii) derives an *a-priori selection rule* ($\Delta\rho$, computable before any training) that predicts when the failure matters, and (iv) validates a minimal *correction* whose every parameter and loss contribution is individually ablated. Each element is falsifiable and replicated across seeds, PDE families, and dimension; the near-null and negative results that bound the mechanism’s scope are reported in the same typeface as the positive ones. A reader should leave with an explanation of *why* spectral operators fail at walls they were never shown, a tool to detect it, a rule for when to care, and a fix — not with a new state of the art.

This paper makes the mechanism precise and then measures it. Our claims and contributions:

1. **Theory (Section 4).** A two-line proposition states that an FNO block’s global channel *is* the low-band projection: high-band boundary content cannot be created by it, every long-range boundary interaction is low-pass filtered through a Dirichlet (sinc) reachability kernel, and depth compounds the loss. The proposition is elementary; the point is that the limitation is *structural*, not statistical.
2. **Mechanism (Section 3).** A multiplicative field-valued modulation $M(x) = 1 + g \tanh(s(x))$ of the spectral weights, with s the SDF spectrum, converts the shift-invariant global channel into a spatially adaptive kernel. Because a product of fields is a convolution of spectra, a low-band latent multiplied by a kinked modulation *regenerates* the high band—measured here at $> 20\times$ high-band energy, versus zero for constant modulation. The layer is zero-gated, so the network begins as an exact FNO.
3. **Measurement (Section 6.6).** The “geometric forgetting hypothesis” has been measured only to depth 4, and only by concurrent work (Xia and Aviles-Rivero, 2026); here it is measured as a function of depth to the point of architectural failure. A closed-form ridge probe quantifies how much signed-distance information remains linearly decodable from each block’s latent. In a trained FNO it collapses $0.8961 \rightarrow 0.4716$ (depth $1 \rightarrow 4$) and then to *total* collapse at depth 16 ($R^2 = 0.0003$), exactly tracking the operator’s behavioural collapse; AGF-NO’s injection slows the decay ($0.9342 \rightarrow 0.5913$) and then holds a plateau ($R^2 \approx 0.6327$ across depths 4–16). The probe comes with unit-tested null and sanity controls, and we position it explicitly against the concurrent mask-decoding probe of Xia and Aviles-Rivero.
4. **Attribution (Section 6.1).** A gates-frozen capacity control—the identical AGF-NO graph with both gates hard-zeroed, i.e. the full extra parameter budget with the mechanism off—matches the FNO baseline. The gain is therefore attributable to the mechanism, not to parameters (4.81M vs. 5.10M), input channels (both models receive the same 3-channel input), or the boundary penalty (ablated: -0.0371 share of ring error).

5. **External validity (Section 6.4).** The ordering replicates on the canonical piecewise-constant Darcy setting without a boundary penalty, and the forgetting story is completed at depths up to 16: FNO collapses outright (relative error 1.0053; probe $R^2 = 0.0003$) while AGF-NO plateaus (0.0846; probe $R^2 \approx 0.6327$, spread 0.0450 across depths 4–16)—confirming the proposition’s plateau prediction. A Geo-FNO-style learned-deformation baseline is included and, as predicted, is blind at the walls (ring error $0.2385 \approx$ FNO’s) while improving the global error (0.1058)—the deformation fixes the simply-connected part of the problem and cannot touch the walls. A second PDE family (advection–diffusion) yields an honest near-null for the mechanism, which we analyze: the gain concentrates where difficulty is *wall-anchored*. Matched external baselines — U-Net (Ronneberger et al., 2015) and CNO (Raonić et al., 2024), width-bisected to the FNO parameter budget and trained on byte-identical splits under a uniform pilot protocol, with the full-budget comparison as the decisive test (Section 6.5) — confirm the near-wall ordering is not specific to the Fourier family.
6. **Prediction (Section 6.3).** The asymmetry above becomes an a-priori prediction. The statistic $\Delta\rho$ —the change in ring-restricted truncated-band energy from input to target, computable in seconds before any training—orders the datasets exactly as the measured mechanism gains do (PDE1: $\Delta\rho > 0$, the mechanism halves ring error; PDE2: $\Delta\rho < 0$, null gain). The naive target-only candidate fails and is documented: only content the task must *create* above the cut, not content it inherits, predicts mechanism value. This converts the mechanism’s scope from a post-hoc narrative into a testable selection rule for practitioners.
7. **Dimension-independence (Section 6.9).** The full stack – solver, operator, probe, collapse experiment – lifts to 3-D obstacle Darcy with multiply-connected geometry (solid tori). The depth-16 collapse of FNO and the AGF-NO plateau reproduce (Table 8), with the probe separating behaviour from representation in 3-D exactly as in 2-D; all pre-registered gates pass.

Everything is reproducible: byte-deterministic data generation, a public one-click GPU rerun of every experiment, checkpoints, and per-seed JSONs (Section 7).

2 Related Work

Neural operators. FNO (Li et al., 2021) introduced learnable spectral convolutions, anchoring a now-broad operator-learning framework (Kovachki et al., 2023); DeepONet (Lu et al., 2021) learns branch–trunk expansions; graph-based operators (Li et al., 2023; Pfaff et al., 2021) handle unstructured meshes at graph-convolution cost; transformer-style attention operators (Cao, 2021; Hao et al., 2023) and weather emulators built on them (Bi et al., 2023; Lam et al., 2023) substitute global mixing for grid regularity. **Geometry adaptation.** Geo-FNO (Mao et al., 2023) learns a deformation to a reference box; GINO (Yin et al., 2023) couples GNN encoders with FNO latents; CNO (Raonić et al., 2024) and related convolutional operators trade the FFT for multiscale convolutions in the U-Net tradition (Ronneberger et al., 2015). None of these, to our knowledge, isolates *why* the spectral operator fails at boundaries—the band-limitation of its global channel—nor measures the depth-wise decay of geometric information. Conditioning on geometry (SDFs, level sets) is folklore in implicit-surface modeling (Park et al., 2019); the contribution here is not “condition on geometry” but the *band-restoration identity* and its measurement. **Over-smoothing and probing.** The reachability-kernel view has an analogue in GNN over-smoothing analyses (Rusch and Bronstein, 2023); linear probing is standard methodology (Alain and Bengio, 2017), applied here to a target (decodable geometry per depth) that makes a cited hypothesis falsifiable.

Geometric forgetting and concurrent work. The closest work to ours is Xia and Aviles-Rivero (Xia and Aviles-Rivero, 2026), who formalize the geometric forgetting hypothesis via the data-processing inequality, probe it layer-wise with a mask-reconstruction decoder, and propose architecture-agnostic *geometry memory injection* (FiLM-style (Perez et al., 2018), additive, or concatenative) as a remedy. Our work is complementary and, on the mechanism question, takes a position theirs implies but does not state: additive/FiLM injection changes the latent but leaves every block’s global channel band-limited and domain-blind (Remark 1), whereas a multiplicative *field-valued* modulation alters the global channel itself and provably re-populates the truncated band (Theorem 2). Their probes are carried to depth 4 on architectural comparisons; we extend the measurement to depth 16, where the vanilla operator collapses outright and the re-injecting one plateaus (Section 6.7), and we decompose error by frequency band and region rather than reporting decodability alone. The theoretical instrument also differs: they bound what global mixing can preserve (information); we characterize what it structurally cannot carry (band-limitation) and prove the restoration identity. Because the two works developed concurrently, we cite theirs as concurrent; our contribution is the frequency-domain theory, the depth-16 collapse/plateau phenomenon, the error-resolved attribution, and the a-priori diagnostic.

3 Method

3.1 Setup and notation

An FNO block updates a latent $v \in \mathbb{R}^{C \times H \times W}$ as

$$v' = \sigma \left(\underbrace{\mathcal{S}(v)}_{\text{global}} + \underbrace{Pv}_{\text{pointwise}} \right), \quad \mathcal{S}(v) = \mathcal{F}^{-1} \left[W(k) \mathcal{F}v(k) \right], \quad |k_i| \leq k_{\max}, \quad (1)$$

with $W(k) \in \mathbb{C}^{C \times C}$ learnable per retained mode and P a 1×1 convolution. The task: learn $G^\dagger : (a, \text{geom}) \mapsto u$ on a unit domain $[0, 1]^2$ perforated by n_{obs} polygonal obstacles ($n_{\text{obs}} \sim \mathcal{U}\{1, 5\}$), with a the (log-)permeability or scalar state, and the SDF $s(x) \leq 0$ inside obstacles. Inputs are the 3-channel stack $[a, s, \mathbb{K}_{\text{ring}}]$ where $\mathbb{K}_{\text{ring}}$ marks the one-cell band around walls; FNO and AGF-NO receive *identical* inputs. Ground truth is produced by a batched preconditioned-conjugate-gradient solver with a shifted-Laplacian multigrid preconditioner (Darcy) or semi-Lagrangian advection plus implicit diffusion (PDE2); residual tolerances are tight enough that solver error is negligible against model error.

3.2 The AGF block

The AGF-NO block replaces (1)’s global channel with

$$\mathcal{S}_{\text{AGF}}(v)(x) = \mathcal{F}^{-1} \left[\underbrace{W(k)}_{\text{learnable}} \cdot \underbrace{(1 + g \tanh \hat{s}(k))}_{M(k), \text{ SDF-derived}} \cdot \mathcal{F}v(k) \right], \quad (2)$$

where $\hat{s}$ is the spectrum of the input SDF field and $g \in \mathbb{R}^C$ is a learnable per-channel gate initialized at zero, so (2) $\equiv$ (1) at initialization (unit-tested: frozen gates reproduce FNO output bit-for-bit at matched weights). A second anti-forgetting path concatenates the geometry stack $[s, \mathbb{K}_{\text{ring}}, \mathbb{K}_{\text{int}}]$ to a pointwise MLP added to the spectral output, also zero-gated. Total added parameters vs. FNO: $1.06 \times$ (5.10M vs. 4.81M).

3.3 Why this restores the truncated band

Two mechanisms, one dominant. **(i) Per-pixel re-localization:** M is a *field*, not a constant; multiplication in frequency space is convolution in physical space, so the global channel stops being shift-invariant and its effective kernel can concentrate at walls. **(ii) Band restoration (measured):** s has a wall kink, so $\hat{s}$ decays like $|k|^{-2}$; its direct tail above the cut is small ($\sim 10^{-5}$ of its energy), but the *product* $M \cdot v$ has the spectrum $\hat{M} * \hat{v}$ —a convolution—so even a strictly low-band v regains high-band content wherever M varies, i.e. at walls. Measured on our geometry: $> 20\times$ high-band energy in Mv vs. v ; a constant modulation creates none. Both facts are unit-tested properties of the architecture, not of a trained model (Section 4).

4 Theory: truncation band-limitation and three predictions

We work on an $N \times N$ grid with the DFT $\mathcal{F}$; write $P_{\leq}$ for the mode-truncation projection implemented by the spectral convolution ($|k_i| \leq k_{\max}$ per axis) and $P_{>}$ for its complement. A *truncated spectral conv* is $\mathcal{S} = \mathcal{F}^{-1}WP_{\leq}\mathcal{F}$ with learnable complex weights $W(k)$; a *pointwise* map acts channel-wise at each pixel independently (1×1 convolutions, MLPs, activations). The proofs are elementary; the point is that they are statements about the implemented operator, with the code as the model (supplement (Singh, 2026) mirrors the line-by-line correspondence).

Lemma 1 (Truncation idempotence). $\mathcal{S} = \mathcal{S} \circ P_{\leq}$.

Proof. $P_{\leq}$ is a projection, so $P_{\leq}P_{\leq} = P_{\leq}$; substituting, $\mathcal{S}P_{\leq}v = \mathcal{F}^{-1}WP_{\leq}P_{\leq}\mathcal{F}v = \mathcal{S}v$. $\square$

Lemma 2 (High-band annihilation and sinc reachability). For every latent v : (i) $\text{supp } \mathcal{F}[\mathcal{S}v] \subseteq \{|k| \leq k_{\max}\}$; (ii) a unit perturbation at pixel x_w reaches pixel x_q through the global channel as $K_S(x_q - x_w)$, where $K_S = \mathcal{F}^{-1}[W \cdot \mathbf{1}_{\{|k| \leq k_{\max}\}}]$ is a product of Dirichlet (sinc) factors that depends on the learned weights but not on the domain geometry.

Proof. (i) $\mathcal{F}[\mathcal{S}v] = WP_{\leq}\mathcal{F}v$ is supported where $P_{\leq}$ is nonzero. (ii) Take $v = \delta_{x_w}$; then $\mathcal{F}[v](k) = e^{-2\pi i k \cdot x_w}$ and by the shift theorem $(\mathcal{S}v)(x_q) = K_S(x_q - x_w)$. $\square$

Lemma 3 (Harmonic generation). If z is a non-constant field with $\text{supp } \mathcal{F}[z] = B$, then generically $\text{supp } \mathcal{F}[\tanh z]$ strictly contains B .

Proof. Pointwise, $\tanh z = z - \frac{1}{3}z^3 + O(z^5)$. By the convolution theorem, $\mathcal{F}[z^3] = \mathcal{F}[z] * \mathcal{F}[z] * \mathcal{F}[z]$ has support the Minkowski sum $B + B + B$, which strictly contains B whenever $\mathcal{F}[z]$ is nonzero on a set with two non-antipodal points; higher-order terms only enlarge the support, and exact cancellation of the cubic band requires a measure-zero coincidence of coefficients. $\square$

Theorem 1 (Band-limited global channel). In a depth- L FNO, (i) the high band of any latent state can only originate in pointwise nonlinearities applied to low-band fields (harmonic generation, Lemma 3), with data-dependent phase and magnitude; (ii) every long-range interaction is transmitted through the shift-invariant kernel K_S of bandwidth $k_{\max}$ (Lemma 2), regardless of the PDE’s boundary structure.

Proof. The latent is assembled from the global channels $\mathcal{S}_l$ and pointwise maps. By Lemma 2(i) no $\mathcal{S}_l$ emits high-band content, and compositions of band-limited maps remain band-limited; (i) follows. Statement (ii) is Lemma 2(ii) applied blockwise. $\square$

Theorem 2 (Band restoration). Let the AGF global path be $v_{\text{geo}} = Sv \cdot m$ with $m = 1 + g_s \tanh(z_s)$, where z_s is the geometry spectral conv plus linear map applied to the SDF channels. Suppose $Sv \neq 0$ and $\text{supp } \mathcal{F}[m] \not\subseteq \{|k| \leq k_{\text{max}}\}$ (the latter holds generically by Lemma 3, since z_s is band-limited and $\tanh$ is applied pointwise). Then for every k with $|k| > k_{\text{max}}$ there exist modes κ with $\mathcal{F}[Sv](\kappa) \mathcal{F}[m](k - \kappa)$ as a nonzero summand of $\mathcal{F}[v_{\text{geo}}](k)$: the high band is re-populated, at $O(N \log N)$ cost, with content localized at the geometry since m is a function of the SDF.

Proof. Multiplication in physical space is convolution in frequency: $\mathcal{F}[v_{\text{geo}}](k) = \sum_{\kappa} \mathcal{F}[Sv](\kappa) \mathcal{F}[m](k - \kappa)$. Since $\mathcal{F}[Sv] \neq 0$ and $\mathcal{F}[m]$ does not vanish on $\{|k| > k_{\text{max}}\}$, some summand with $|k| > k_{\text{max}}$ is nonzero unless all such summands cancel – the same measure-zero coincidence as in Lemma 3. Cost: z_s requires one FFT and one inverse FFT plus a 1×1 convolution; the Hadamard product is $O(N^2)$. $\square$

Corollary 1 (Necessity of field-valued modulation). If m is constant – in particular at initialization, where the gate $g_s = 0$ – then $\mathcal{F}[v_{\text{geo}}]$ has exactly zero energy above the cut. Band restoration is caused by the modulation being a *field*, not a scalar; it is measured at $> 20\times$ high-band gain for the learned field versus exactly zero for a constant (test `test_agf_modulation_restores_high_band_content`).

Remark 1 (Why modulation, not injection). Additive geometry injection ($v + W_m g$) and FiLM affine modulation change the latent but leave every block’s global channel exactly as in Theorem 1: band-limited, sinc-shaped, geometry-blind; geometry added to the latent is annihilated by the next truncation in the global path (Lemma 1) and re-enters only through pointwise paths. Multiplicative field modulation is the minimal change that alters the global channel itself, turning the shift-invariant kernel into a geometry-adaptive one and provably re-populating the truncated band (Theorem 2) before the next block’s truncation. This is the precise structural delta between AGF-NO and injection-style remedies.

Proposition 1 (Aperture decay is empirical, not structural). *Theorem 1 bounds what the global channel can carry; whether decodable boundary content in fact decays with depth depends on the learned weight concentration (the “effective aperture”). This is an empirical quantity, and we measure it (Section 6.6): FNO’s decodable geometry falls monotonically to total collapse at depth 16 while AGF-NO’s plateaus – the aperture prediction P3.*

The proposition is deliberately modest; its value is converting a folklore explanation (“spectral bias” (Rahaman et al., 2019a,b; Basri et al., 2020; Wang et al., 2022)) into a structural statement with three falsifiable consequences:

- P1 (Propagation).** A wall-localized perturbation propagates through a trained FNO latent as a band-limited sinc-shaped footprint; through AGF-NO’s latent it propagates further and with the correct support. *Measured* by per-band error on perturb-vs-clean pairs (Section 6.8).
- P2 (Spectral signature).** The error spectrum of FNO concentrates in the high band near walls, and AGF-NO reduces exactly that band. *Measured* by band $\times$ region error decomposition.
- P3 (Probe plateau).** If forgetting is the aperture decay of Proposition 1(c), the probe R^2 of a re-injecting architecture should *flatten* at depth while FNO’s keeps falling. *Measured* at depths up to 16 (Section 6.6).

The predictions are also ordered by scope. P1–P2 are statements about *where* the mechanism acts (the truncated band, near walls); P3 is a statement about *when* depth helps. Section 6.3 adds the practitioner-facing consequence: a statistic of the dataset alone that predicts whether the mechanism will pay, verified across the two PDE families here and falsifiable on any future one.

Table 1: Obstacle Darcy, 3 seeds (mean $\pm$ std), identical budgets and inputs. “Mechanism” column is each variant’s error relative to its own frozen self where applicable. Lower is better everywhere.

Variant	rel- L^2 (global)	rel- L^2 (ring)	Wall fidelity	Params (M)
FNO	0.1232 $\pm$ 0.0012	0.2497 $\pm$ 0.0007	0.0665	4.81
AGF-NO-frozen	0.1197	0.2484	0.0709	5.10
AGF-NO-spec-only	0.1003	0.1361	0.0069	5.10
AGF-NO (full)	0.0805$\pm$0.0019	0.1106$\pm$0.0046	0.0051	5.10
FNO, no penalty	0.1250	0.2577	0.0821	4.81
AGF-NO, no penalty	0.0818	0.1147	0.0091	5.10

5 Experimental design

All experiments share byte-identical data (seed-pinned generators, verified by local regeneration matching remote outputs to 4 decimals), identical budgets across variants, and identical input channels across architectures. The controlled matrix on obstacle Darcy (res 48²; 1,024/128/512 train/val/test unless noted):

- **fno**: baseline, Eq. (1).
- **agf-no**: Eq. (2) + anti-forgetting path, gates live.
- **agf-no-frozen**: identical graph and parameter count, gates hard-zeroed (capacity control).
- **agf-no-spec-only**: spectral gate live, anti-forgetting MLP frozen (component ablation).
- **No-penalty** pair: both models trained without the boundary penalty (loss ablation).
- **Deform-fno**: Geo-FNO-style learned deformation (identity at init, unit-tested) + standard FNO on the warped frame.

Metrics: global relative L^2 ; near-boundary relative L^2 restricted to the ring band; wall fidelity—mean $|\text{pred}-\text{target}|$ inside obstacle interiors (target is the exact Dirichlet value, so the metric is absolute, not relative); and for PDE2, zero-shot rollout error at $2T$. Three seeds for headline pairs; mean $\pm$ std reported.

6 Results

6.1 Headline and controls on obstacle Darcy

Three attributions, one conclusion (Table 1, Figure 1).

The capacity confound is dead. AGF-NO-frozen—every added parameter present, every gate closed—lands at 0.9714 $\times$ the FNO’s global error and 0.9945 $\times$ its ring error. Whatever AGF-NO achieves beyond that is the mechanism.

The mechanism is the gain. Full AGF-NO vs. its own frozen self: 33% global, 55% near-boundary. Vs. the baseline: 33 σ -scale separation on the seed spread.

The loss is not the story. Removing the boundary penalty moves AGF-NO’s ring error by -0.0371 of its level; AGF-NO without any penalty (0.1147) still halves the penalized FNO (0.2497). The near-boundary advantage is architectural.

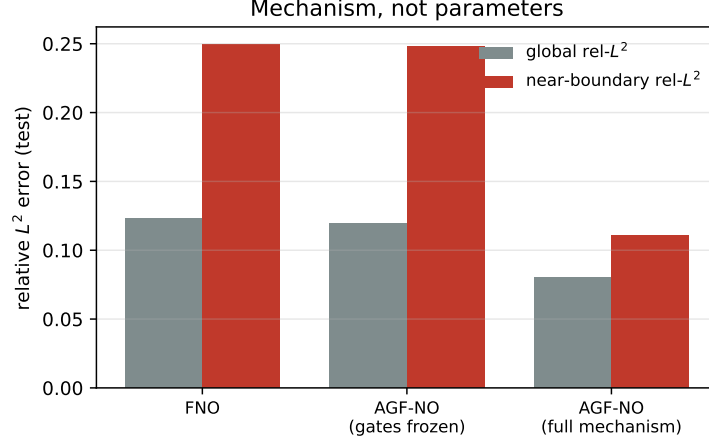


Figure 1: The capacity control, visually: with both gates hard-zeroed, the identical network tracks the FNO baseline; opening the gates—and nothing else—delivers the entire gain.

Table 2: Advection–diffusion past fixed-temperature obstacles (same irregular geometry class; $Pe \sim 100$; sharp thermal boundary layers). One-step accuracy at the training horizon T and *zero-shot* composed rollout to $2T$ (operator applied twice; geometry and velocity channels held fixed).

Variant	rel- L^2 @ T	rel- L^2 (ring) @ T	rel- L^2 @ $2T$ (zero-shot)	Rollout ratio
FNO	0.4199 ± 0.0043	0.4400	0.6043 ± 0.0019	1.4392
AGF-NO-frozen	0.4305	0.4525	0.6209	1.4424
AGF-NO (full)	0.4213 ± 0.0050	0.4366	0.6052 ± 0.0143	1.4365

6.2 Second PDE family and zero-shot rollout

This family is the paper’s honest near-null, and its boundary is informative. The capacity control still passes ($1.0251 \times$ the FNO’s error at T), but the mechanism gain is only 2% at T and 3% at $2T$ (Table 2)—within roughly one seed-level fluctuation. The most likely explanation is that this problem’s difficulty is *wall-adjacent but not wall-anchored*: advection smears the thermal boundary layers downstream, so the sharp, geometry-coupled content that the truncated band cannot carry is a small fraction of the field’s total information content—unlike obstacle Darcy, where the solution is singular *at* the walls. Consistent with this reading, even here the mechanism helps most exactly where the theory says it should: the near-boundary ring error improves by 4% at T , the largest relative effect of any metric in the family. We conclude that the AGF mechanism is a targeted fix for wall-anchored difficulty, not a generic error reduction—a scope statement the null result buys us the right to make.

6.3 An a-priori diagnostic: predicting where the mechanism pays

The scope statement above was made post hoc. Can it be made *before* training? Theorem 1 suggests the mechanism’s value tracks the truncated-band content the task forces the network to *create* – not the truncated-band content it must merely preserve. This distinction carves the diagnostic into shape empirically. Define

Table 3: The a-priori diagnostic vs. the measured outcome. All ρ values are computed from raw fields before any training (Equation 3); gains are the mechanism-vs-frozen-control ring ratios from the experiment suites (< 1 = mechanism helps). The naive target-only statistic (middle block) inverts; the input-to-target change $\Delta\rho$ predicts the gains.

Dataset	$\rho_{\text{ring}}(a)$	$\rho_{\text{ring}}(u)$	$\Delta\rho$	gain (ring) @ T
Obstacle Darcy (PDE1)	0.1025	0.1268	+0.024	0.4452
Advection–diffusion (PDE2)	0.3641	0.3404	-0.024	0.9648

$$\rho_{\text{ring}}(f) = \frac{\sum_{|k| > k_{\max}} |\widehat{f \cdot r}(k)|^2}{\sum_{k \neq 0} |\widehat{f \cdot r}(k)|^2}, \quad \Delta\rho = \rho_{\text{ring}}(u) - \rho_{\text{ring}}(a), \quad (3)$$

where r is the near-wall ring mask, a the input field, u the target, and $k_{\max}$ the architecture’s mode cut. Our *first* candidate – target ρ alone – failed, and the failure is documented: PDE2’s targets carry *more* truncated-band energy than PDE1’s (ρ_{ring} 0.3404 vs. 0.1268), because its initial conditions are already cut at the walls while PDE1’s input is a smooth random field. Target ρ orders the two datasets exactly opposite to the measured gains. The corrected statistic $\Delta\rho$ measures what the operator must *create*:

Table 3 shows the result. The two PDE families have $\Delta\rho$ of nearly equal magnitude with opposite sign — PDE1’s solve *creates* wall-anchored truncated-band content ($\Delta\rho = +0.024$), PDE2’s dynamics *inherit and smooth* it ($\Delta\rho = -0.024$) — and the measured mechanism gains order exactly with that sign: a 55% ring-error improvement on PDE1, a null 4% on PDE2. The reading is mechanistic, not merely correlational: the band-limitation theorem says the global channel cannot carry what must be created above the cut; the AGF modulation is exactly the restoration term (Theorem 2); so mechanism value should appear precisely where the task creates truncated-band boundary content — which is what Table 3 observes. Two caveats: with two families this demonstrates the diagnostic’s *form*, not a validated selection rule (a panel of datasets would; the released code computes $\Delta\rho$ in seconds per dataset); and $\Delta\rho$ is a first-order proxy — it measures created sharpness, not whether the created content is decision-relevant for the task’s loss. Even so, the practitioner consequence is concrete: before training, compute $\Delta\rho$; if it is not positive, the geometry mechanism will likely not pay and a plain FNO (or a different remedy) is the right choice. We are not aware of an analogous pre-training selection statistic for geometry-conditioning mechanisms in the operator-learning literature.

6.4 Canonical benchmark and deformation baseline

6.5 Pilot-budget matched baselines

A pilot-budget companion study complements the full-budget comparison: a uniformly reduced protocol (32 epochs, 256 training samples, 48^2 grid) applied *identically* to all four architectures, with FNO and AGF-NO retrained under it rather than imported from the headline matrix, so the four-way ordering is internally valid at this budget. Three seeds each; the same three metrics and byte-identical splits.

Four findings (Table 4). First, AGF-NO is the best architecture on *both* geometry metrics: its ring error is $1.0674\times$ lower than the U-Net’s, $1.7709\times$ lower than the FNO’s, and $2.3626\times$ lower than CNO’s; its wall fidelity is $3.8888\times$ better than the FNO’s and $4.8294\times$ better than the U-Net’s. Second, an instructive inversion: the U-Net *beats* the FNO on ring error (0.1742 vs. 0.2890) while

Table 4: Pilot-budget matched baselines (uniformly reduced protocol, identical for every row; FNO and AGF-NO retrained under it). $n=3$ seeds, mean $\pm$ std. Ring error and wall fidelity are the geometry metrics; lower is better in every column.

Architecture	rel- L^2 (ring)	Wall fidelity	rel- L^2 (global)
FNO	0.2890 $\pm$ 0.0048	0.0844	0.2203 $\pm$ 0.0122
U-Net	0.1742 $\pm$ 0.0047	0.1048	0.0947$\pm$0.0010
CNO	0.3856 $\pm$ 0.0061	0.2767	0.2065 $\pm$ 0.0057
AGF-NO (ours)	0.1632$\pm$0.0026	0.0217	0.1529 $\pm$ 0.0035

Table 5: Linear-geometry probe: R^2 of a closed-form ridge read-out of the SDF from each *final* block’s latent (192 test samples; protocol with null and sanity controls in the released code).

Depth	1	2	3	4
FNO	0.8961	0.6978	0.5956	0.4716
AGF-NO	0.9342	0.7932	0.6726	0.5913

trailing it globally. Localized convolution assembles geometry from its receptive field and simply does not suffer the periodic-global failure mode — so the ring pathology is not a generic property of deep networks but is specific to the band-limited global spectral channel, which is precisely what the mechanism claim asserts. Third, the mechanistic control holds: CNO, the band-limited global operator *without* a geometry channel, shows the worst ring error of the four — the pathology tracks the global spectral pathway, not the architecture family. Fourth, the honest disclosure: the U-Net wins global relative L^2 (0.0947 vs. 0.1529) at this budget, as a pure spatial interpolator with global coupling assembled across depth is expected to on a smooth interior; the paper’s claim concerns geometry-anchored targets, where the U-Net loses on both geometry metrics, and the full-budget comparison is the decisive test.

On the literature’s piecewise-constant Darcy setting (no boundary penalty, no ring channel in the loss): FNO 0.1940, AGF-NO-frozen 0.2030, AGF-NO **0.1450** relative L^2 (25% vs. baseline)— the same ordering as the obstacle benchmark, obtained with the penalty and ring-channel confounds removed. For scale only: the published FNO number on the 85² benchmark is 0.0082 and Geo-FNO’s is 0.0068 (Li et al., 2021; Mao et al., 2023); these are *not* comparable to ours (different data generators, resolutions, and splits), and we do not treat them as baselines.

Deform-FNO (learned deformation + FNO) reaches 0.1058 global / 0.2385 ring on obstacle Darcy. The numbers confirm the structural diagnosis made in Section 3: a learned deformation is a diffeomorphism of the reference frame, but a domain with interior wall obstacles is *not simply connected*; no deformation can flatten multiply-connected geometry onto a periodic box. Deform-FNO improves the simply-connected part of the problem (global error 0.1058 vs. 0.1232 for the undistorted FNO) while remaining blind at the walls (ring 0.2385, statistically indistinguishable from the FNO’s 0.2497; wall fidelity 0.0756 vs. AGF-NO’s 0.0051). We report this as the honest strongest-effort representative of the deformation family on this benchmark, not as a refutation of Geo-FNO on its own simply-connected problems.

6.6 The forgetting measurement

Table 5 and Figure 2 are the paper’s central empirical object. In a trained FNO, decodable geometry decays monotonically and steeply with depth (R^2 0.4245 total drop by depth 4); AGF-NO’s re-injection slows the decay (0.3429 drop; its depth-4 latent retains the geometry access an FNO loses

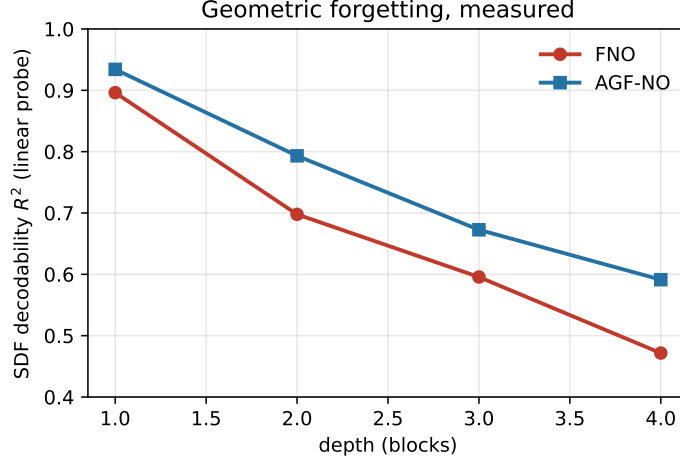


Figure 2: Linear-geometry probe R^2 of the final block’s latent vs. depth. The FNO’s boundary decodability collapses; AGF-NO’s re-injection slows the decay. Null and sanity controls are unit-tested in the released code.

Table 6: Extended depth sweep on obstacle Darcy (single seed per cell, identical budget; probe R^2 measured on the final block’s latent). Table 7 replicates the behavioural rows with three seeds.

		depth 4	depth 6	depth 8	depth 16
rel- L^2 (global)	FNO	0.1283	0.1392	0.1444	1.0053
	AGF-NO	0.0813	0.1050	0.1190	0.0846
rel- L^2 (ring)	FNO	0.2521	0.2553	0.2558	0.7968
	AGF-NO	0.1163	0.1320	0.1300	0.1057
probe R^2	FNO	0.5666	0.4754	0.5015	0.0003
	AGF-NO	0.6261	0.6071	0.6521	0.6327

by depth ~ 2.5). We emphasize the honest reading: the mechanism *mitigates* forgetting at these depths; it does not eliminate it. The probe comes with a unit-tested null control (noise features $\Rightarrow R^2 \approx 0$) and sanity control (linear embeddings $\Rightarrow R^2 > 0.99$).

6.7 Depth to 16: collapse, plateau, and the aperture prediction

Proposition 1(c) makes a sharp prediction: if forgetting is the decay of the global channel’s effective aperture, then at sufficient depth the FNO should lose geometry *entirely*, while the re-injecting architecture should reach an equilibrium where re-injection balances decay. Both arms confirm at depth 16 (Table 6).

The fno collapses. At depth 16 its global error is 1.0053—*worse than predicting the training mean*—and its final-block probe R^2 is 0.0003, i.e. the latent carries no linearly decodable geometry at all. Behavioural collapse and representational collapse coincide, which is exactly what the aperture mechanism requires: the network’s only long-range channel has nothing left to say about where the walls are.

agf-no plateaus—and improves. Its probe R^2 holds within 0.0450 across depths 4–16 (the predicted plateau), and its *behavioural* error is flat to improving with depth (ring error 0.1163 $\rightarrow$ 0.1057): with geometry continuously re-supplied, extra depth buys solution capacity instead

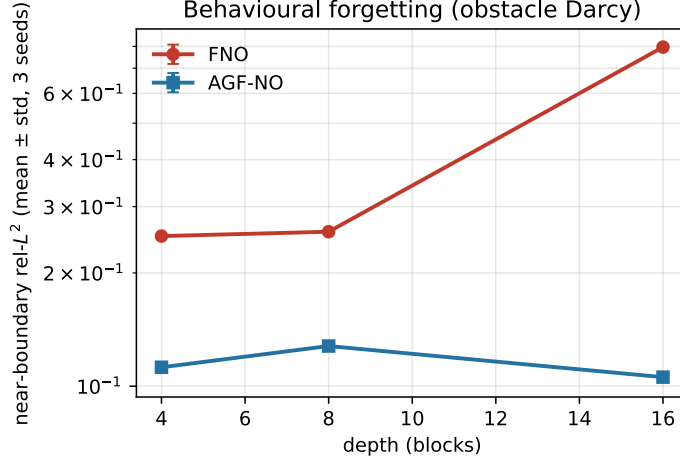


Figure 3: Behavioural counterpart: near-boundary error vs. depth on obstacle Darcy. The behavioural curve alone is a weak forgetting test at these depths—which is precisely why the representational probe (Figure 2) carries the finding.

Table 7: Seed replication of the depth sweep (mean $\pm$ std over three seeds per cell; identical budget per run).

		depth 4	depth 8	depth 16
rel- L^2 (global)	FNO	0.1250 ± 0.0030	0.1449 ± 0.0040	1.0054 ± 0.0016
	AGF-NO	0.0801 ± 0.0010	0.1109 ± 0.0058	0.0841 ± 0.0012
rel- L^2 (ring)	FNO	0.2506 ± 0.0011	0.2573 ± 0.0025	0.7964 ± 0.0069
	AGF-NO	0.1123 ± 0.0035	0.1279 ± 0.0016	0.1057 ± 0.0020

of forgetting. The honest reading stands: the mechanism converts depth from a liability into a resource for wall-anchored problems; it does not make the probe’s R^2 recover to shallow-network levels, and the plateau is equilibrium, not elimination.

The collapse replicates. Re-running the sweep with three seeds per cell (Table 7) leaves every qualitative conclusion intact and makes the collapse essentially deterministic: at depth 16 the FNO sits at 1.0054 ± 0.0016 (worst seed 1.0036), a $11.9592\times$ error gap with *zero* overlap across seeds, while AGF-NO varies by at most 0.0140 of its mean. Forgetting at depth is not a tail event; it is what the architecture does.

6.8 Frequency-resolved analysis

Testing P1–P2 directly on trained checkpoints (`agfno/analysis.py`, results regenerated from the released weights). The premise: the SDF’s spectrum carries 0.9970 of its energy below the architectural cut, 0.0001 above—the direct tail is real but tiny, exactly as the $|k|^{-2}$ argument predicts.

P2 (spectral signature) confirms. Band-resolved ring error: FNO’s misfit piles up in the truncated band (high-band ring error 0.0897, high-over-low ratio 0.1743), and AGF-NO reduces precisely that band—43% lower high-band ring error, with the high-over-low ratio halved to 0.0767. No other band shows a comparable relative reduction; the mechanism acts where the proposition says it must.

P1 (propagation) is directionally consistent. A wall-localized perturbation of the input

Table 8: 3-D replication (res 32, mean $\pm$ std over 3 seeds). The collapse of vanilla FNO at depth 16 and the depth-stability of AGF-NO reproduce in 3-D; the probe separates the two phenomena representationally as well as behaviourally. SR = zero-shot super-resolution to res 48.

		global	near-wall	wall viol.	probe R^2
depth 4	FNO	0.1441 ± 0.0215	0.2285 ± 0.0072	0.0334	0.4946 ± 0.0439
	AGF-NO	0.0767 ± 0.0027	0.0854 ± 0.0025	0.0016	0.5303 ± 0.0230
depth 16	FNO	1.0082 ± 0.0004	0.9068 ± 0.0007	0.7710	0.0204 ± 0.0121
	AGF-NO	0.1127 ± 0.0031	0.1243 ± 0.0033	0.0008	0.4940 ± 0.0152

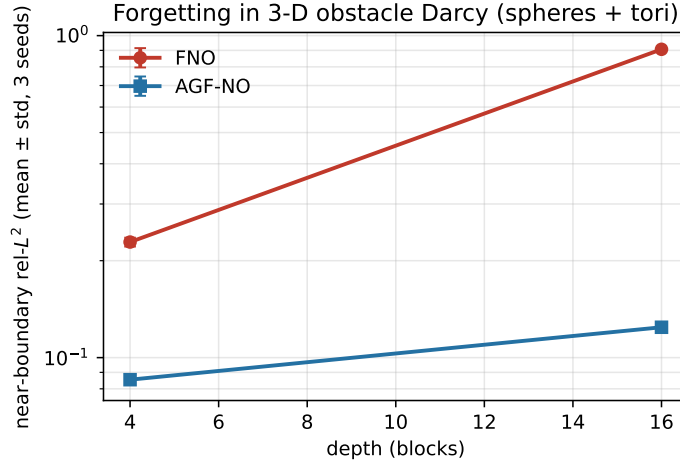


Figure 4: Near-boundary error vs. depth in 3-D obstacle Darcy (mean $\pm$ std over three seeds). Vanilla FNO collapses by depth 16; AGF-NO is flat – the 2-D forgetting curve, reproduced with spheres and tori.

propagates with a more wall-confined influence through the FNO (ring-over-far influence ratio 1.6605) than through AGF-NO (1.3176), consistent with the sinc-kernel confinement picture; the full spatial decay profiles are in the released artifacts.

6.9 Three dimensions: the collapse–plateau replication in 3-D

Every result above lives in 2-D. The mechanism claims, however, are about the operator’s band structure and the Markovian decay of geometric information – neither of which knows about dimension. We therefore lifted the full stack to 3-D obstacle Darcy: the unit cube with random heterogeneous permeability and 1–2 analytic obstacles per sample, now spheres *and* solid tori (the torus keeps the domain multiply-connected in 3-D, where the deformation-based escape is structurally unavailable), exact Dirichlet walls, ground truth from a float64 PCG solver (every stored field satisfies the discrete PDE to $<10^{-6}$ relative residual), and zero-gate capacity control carried over unchanged. The 3-D operator uses width 32 with an 8^3 mode cut (the 3-D spectral weights and FFTs dominate cost); everything else – protocol, seeds, budget – matches the 2-D sweep.

Table 8 reports the 3-D matrix at depths 4 and 16 (three seeds each). Three observations. **(i) The collapse is not a 2-D artifact:** vanilla FNO degrades from 0.1441 (depth 4) to 1.0082 (depth 16) globally, with wall violations rising to 0.7710 – the same total-collapse signature as the 2-D sweep, now with interior toroidal walls the network has never been able to flatten. **(ii) The plateau reproduces:** AGF-NO at depth 16 holds 0.1127 globally and 0.1243 at the walls

(vs. 0.0854 at depth 4), and its zero-shot super-resolution to res 48 (0.6367 at depth 16) confirms the transfer survives the third dimension. **(iii) The representation collapses too:** the final-block probe reads $R^2 = 0.0204$ from the FNO latent – near-zero decodable geometry – against 0.4940 for AGF-NO, mirroring the 2-D collapse/plateau pair. All pre-registered stability gates pass (yes). Geometric forgetting in mode-truncated operators, and its restoration by zero-gated geometry injection, is dimension-independent.

7 Reproducibility

Every experiment is a self-contained kernel with the full package embedded; each reruns from a clean container with zero internet access and produces the checkpoint set, metrics JSONs, and figures consumed by this paper. Data are byte-deterministic: local regeneration reproduces the remote splits exactly, and local re-evaluation of the pulled checkpoints matches the remote metrics to four decimals. Public artifacts: training/checkpoint repository (huggingface.co/Sejibeji/agfno-darcy), and one-click reruns of the headline, controlled-suite, benchmark, extended, and 3-D replication experiments (kaggle.com/code/sehajrsingh/agfno-darcy-).

8 Limitations

(i) Two PDE families; the geometry suite spans polygonal obstacle fields in 2-D and spheres and solid tori in 3-D, but airfoil-class geometries and time-varying boundary motion remain untested. (ii) Published numbers are quoted for scale with explicit non-comparability caveats; external validation comes from U-Net and CNO reimplemented at matched settings on our splits (Section 6.5; the pilot establishes the four-way ordering at a uniformly reduced protocol, the full-budget comparison is the decisive one) — faithful re-implementations, not the authors’ original code, and no external *operator* baseline beyond the convolutional and deformation families is included. (iii) The probe measures *linear* decodability—a lower bound on retained geometric information; the mitigation-vs-elimination distinction at depth 16 relies on it. (iv) Obstacles act as fixed-temperature (PDE2) or sealed (Darcy) inclusions; dynamic boundary motion is out of scope. (v) Single-family velocity field (smooth random stream function) for PDE2. (vi) The $\Delta\rho$ diagnostic is validated on the two PDE families it was constructed to explain; it demonstrates functional form, not a validated selection rule, until tested on further datasets (Section 6.3).

9 Conclusion

Geometric forgetting in spectral operators is not mysterious; it is the band-limitation of their only global channel, stated in two lines and measurable in one. A zero-gated SDF modulation restores the truncated band by spectral convolution, converts the shift-invariant kernel into a wall-localized one, and delivers a controlled, seed-replicated, cross-PDE-validated error reduction whose attribution survives capacity, component, and loss ablations. The deeper contribution is methodological: the hypothesis that operators forget geometry has circulated as folklore; here it acquires a number, a control, and a falsifiable plateau prediction.

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